MTRX 3700 : Mechatronics 3

Embedded systems in Mechatronics

Dr. Viorela ILA

Acknowledgement of Country

I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. We are currently on the land of the Gadigal people of the Eora nation and pay our respects to their Elders, past, present and emerging.

Course Objectives


Augmented Reality


Drive Assist Systems


Drone Vision


Satellite Vision


Medical Vision


Machine Vision


Broadcast


Physical Security


Thermal Vision

The objective of the course is to provide students with the essential skills necessary to design and program embedded systems for mechatronic applications.

Administrative Details

Lecturers : **Dr. Viorela ILA ,**
Lead tutor: **Lewis Maxwell WATTS**

- Lecture Time : Wednesday 10-12

- Laboratory : Thursday 9-12, 12-3, 3-6

Lectures, Assignments and Reading Material will be available on Canvas and Ed Stem

Contact Details :

– E-mail : viorela.ila@sydney.edu.au

Administrative Details

- Lecture Time : Wednesday 10-12

- The Faculty of Engineering has traditionally expected students to attend at least 90% of scheduled teaching activities. This means students should plan to attend the vast majority of both lectures and laboratory sessions, as consistent attendance is essential for developing the knowledge and practical skills required in engineering

- Lectures may include important explanations, demonstrations, announcements, and examples that are not available anywhere else.

Administrative Details

- Laboratory : Thursday 9-12, 12-3, 3-6

- Attendance at scheduled laboratory sessions is expected. The laboratories involve practical hardware activities, demonstrations, and assessment components that cannot be fully replicated remotely.

- Students are responsible for attending their timetabled laboratory session.
- Lab attendance will be recorded, and it will be used as evidence if we need to balance the SparkPlus factors.

EdStem Quiz and Lessons



Lesson 5 - SoCs (3%)

Due: Sun November 3rd, 11:59pm

EdStem Quiz (2%)

Ed Quiz 0.1 - Familiarising with Quartus, Schematics

Available: Tue August 5th, 9:00am
Due: Wed August 13th, 11:59pm

Ed Quiz 0.2 - Digital Logic Revision

Available: Tue August 5th, 9:00am
Due: Wed August 13th, 11:59pm

Administrative Details

- Tutors
  - Lewis Maxwell Watts lwat0045@uni.sydney.edu.au
  - Viet Quang Nguyen vietquang.nguyen@sydney.edu.au
  - Guangze (Daren) Deng guangze.deng@sydney.edu.au

- Lab tech support
  - Zachary Milgate zachary.milgate@sydney.edu.au

Call for Student Representative

- Will act as a communication channel between staff and students, and will provide feedback regarding the course content, resources, delivery and assessments.
- After being appointed as student representative (week 2), they should make sure the other students know who they are.
- Be contactable (for example by email) by other students in the subject and communicate any feedback with the unit coordinators and tutors (every week!)
- They are also invited to participate to the student-staff liaison meeting in Week 7/8
- Please email the unit coordinator if you would like to nominate yourself as a rep

Course Outline

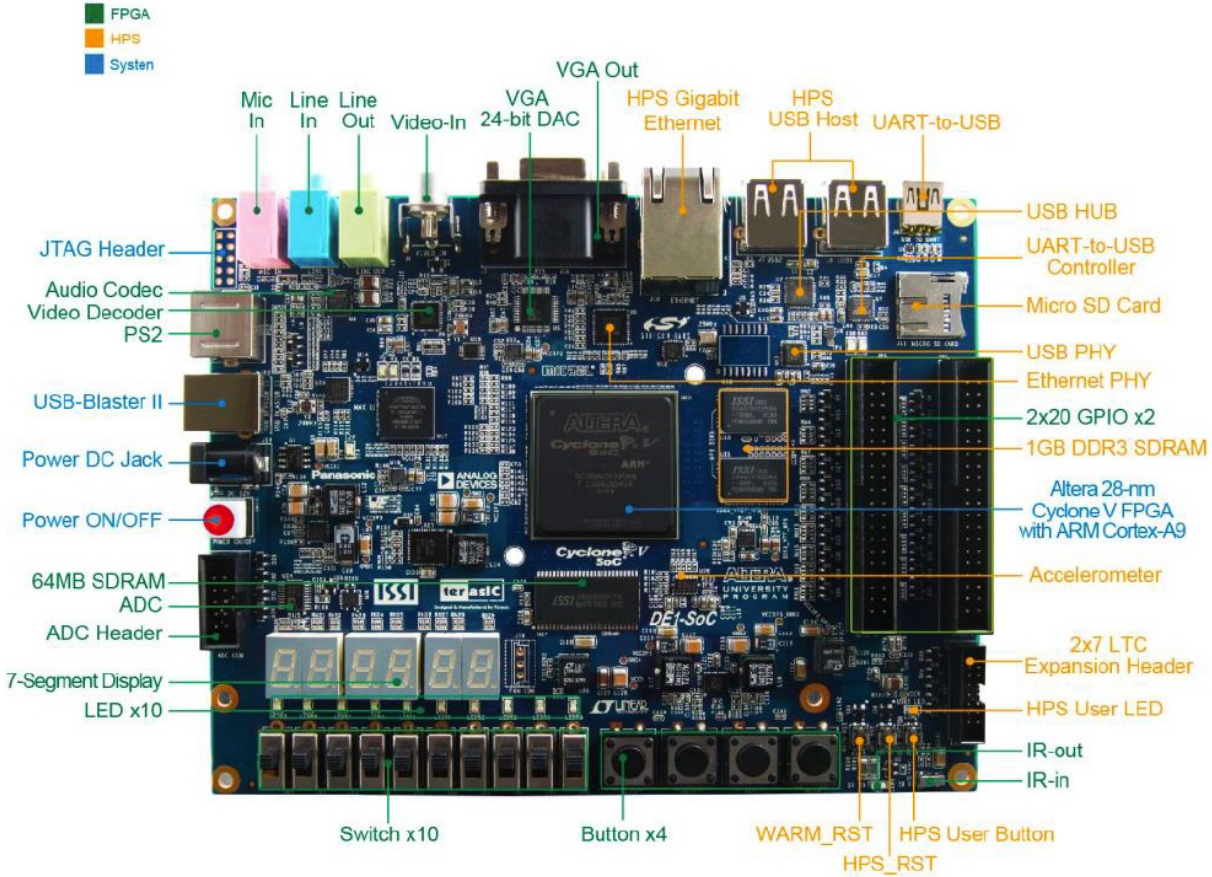
		Lecture	Lab Work	Due
Week 1	Wed 5 Aug 2026	10-12: Introduction to Digital Circuits and FPGA	EQ0.1: Digital Logic	
	Thu 6 Aug 2026	Video: Quartus, Schematic, Simulation	EQ0.2: Install Quartus	
Week 2	Wed 12 Aug 2026	10-12: Introduction to hardware description language		EQ0.1 +EQ0.2
	Thu 13 Aug 2026	Video: First HDL code	Lesson1	
Week 3	Wed 19 Aug 2026	10-12:Modular Design, Simulation & FPGA Hardware		
	Thu 20 Aug 2026	Video: Testbenches	Lesson2	
Week 4	Wed 26 Aug 2026	10-12: FPGA design. Finite State Machine		Lesson 1
	Thu 27 Aug 2026	Video: Synthesis Timing and Optimisation	Assignment 1	
Week 5	Wed 2 Sep 2026	10-12: Communication (wifi, ethernet, serial, parallel)		Lesson 2
	Thu 3 Sep 2026	Video: Communication Examples	Assignment 1 -Demo	Q&A #1 + Assignment 1 -DEMO
Week 6	Wed 9 Sep 2026	10-12: DSP Intro		Assignment 1 - Report
	Thu 10 Sep 2026		Lesson 3: FSM +COMMS	
Week 7	Wed 16 Sep 2026	10-12: Hardware for DSP		
	Thu 17 Sep 2026		Lesson 4 : DSP	
Week 8	Wed 23 Sep 2026	10-12: Review and discussions: First part		Lesson 3
	Thu 24 Sep 2026		Assignment 2	
Week 9	Wed 7 Oct 2026	10-12: Hardware/Software codesign		Lesson 4
	Thu 8 Oct 2026		Assignment 2-Demo	Q&A #2 Assignment 2- Demo
Week 10	Wed 14 Oct 2026		Pitch	Pitch
	Thu 15 Oct 2026			Assignment 2 - Report
Week 11	Wed 21 Oct 2026	10-12: System Design, SoC, soft Processor	MP	
	Thu 22 Oct 2026	Video: Soft Processor Example		
Week 12	Wed 28 Oct 2026	10-12: No lecture		
	Thu 29 Oct 2026	12-13: No lecture	MP	
Week 13	Wed 4 Nov 2026	10-12: Review lecture	Lesson 5 Nios	Lesson 5
	Thu 5 Nov 2026		MP - Demo	Major Project- Q&A + DEMO
	Sun 8 Nov 2026			Major Project - Report

Assessments

EDQuiz	2%	Individual		
ED Lessons	15%	Individual		62% Individual
Q&A	15%	Individual	secured	
DemoA1	4%	Group	secured	38% Group
DemoA2	5%	Group	secured	
DemoMP	8%	Group	secured	65% Secured
Report A1	5%	Group	AI - reference	
Report A2	5%	Group	AI - reference	18% AI
Report MP	8%	Group	AI - reference	
Pitch	3%	Group	secured	
Exam	30%	Individual	secured	

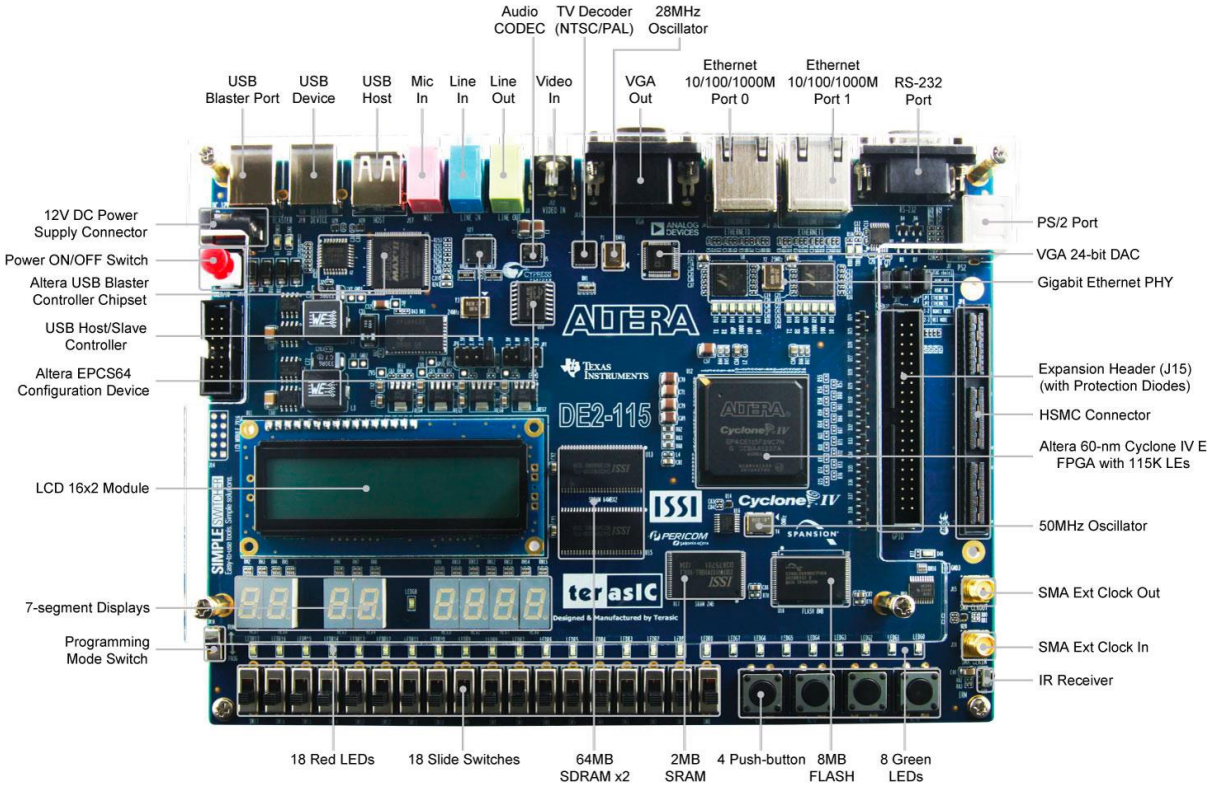
Type	Description	Weight	Due	Length
Assignment	**EDQuiz 0.1 and 0.2**	2%	Week 02	1 week
Assignment	**EDLesson1**	3%	Week 04	2 weeks
Assignment	**EDLesson 2**	3%	Week 05	2 week
	Assignment 1		Week 05	3 weeks
Creative assessment / demonstration	**Demonstration**	4%		
Assignment	**Report**	**5%**	Week 06	
	L1,L2, A1 Q&A	**5%**		
Assignment	**EDLesson 3**	3%	Week 08	2 weeks
Assignment	**EDLesson 4**	3%	Week 09	2 weeks
	Assignment 2		Week 9/10	3 weeks
Creative assessment / demonstration	**Demonstration**	5%		
Assignment	**Report**	5%		
	L3,L4, A2 Q&A	**5%**		
Presentation	**Major Project - Pitch**	3%	Week 10	1 week
Creative assessment / demonstration	**Major Project - Demo**	8%	Week 13	4 weeks
	Major Project - Q&A	**5%**		
Assignment	**Major Project - Report**	8%	Week 13	4 weeks
Assignment	**EDLesson 5**	3%	Week 13	2 weeks
Supervised exam	**Final Exam** Open book	30%	Formal exam period	2 hours

Hardware- DE1-soc board



https://www.terasic.com.tw/cgi-bin/page/archive.pl?Language=English&CategoryNo=167&No=836&PartNo=4#contents

Hardware- DE2-115 board – Available but not the main hardware



https://www.terasic.com.tw/cgi-bin/page/archive.pl?Language=English&CategoryNo=139&No=502&PartNo=4#contents

Hardware

Borrowing the Equipment

1. **Inspect the Equipment**: Check the equipment for any existing damage or missing parts before accepting it. Report any issues immediately.
2. **Use Responsibly**: Handle the equipment with care. Follow any specific usage instructions provided by the institution or manufacturer.
3. **Keep it Safe**: Store the equipment in a secure and appropriate environment to prevent damage, loss, or theft.
4. **Avoid Unauthorized Use**: Do not lend the equipment to others or use it for purposes not related to your project!

Returning the Equipment

1. **Clean and Packed**: Ensure the equipment is clean and properly packed in its original packaging or a secure container.
2. **Check for Damage**: Inspect the equipment for any damage or missing parts. Report any issues when returning it, use technical reports format.
3. **Return on Time**: Adhere to the return deadline to avoid late penalties.

4. **Equipment Return Assessment:** A portion of your assignment mark will be allocated to the condition in which the equipment is returned. Marks will be awarded based on whether the equipment is clean, complete, undamaged, properly packed, and returned on time!

Learning Outcomes (as per course description)

- **LO1**. Understand and plan for the process of incremental implementation, recognising the importance of project management, teamwork, software/hardware co-design, and iterative development by members of a development team.
- **LO2**. Design and prototype the hardware and software comprising a **FPGA-based system**.
- **LO3**. Understand in detail the **hardware architecture** of a modern **FPGA**.
- **LO4**. Understand the basic architecture of traditional and modern **FPGAs and System-on-Chip**
- **LO5**. Understand engineering concepts in the design of digital circuits.
- **LO6**. Understand the role of **hardware description languages** in digital circuit implementation and describe simple hardware functions using a hardware description language.
- **LO7**. Demonstrate the ability to differentiate between CISC, RISC, DSP processors and FPGAs, understanding the reasons for their evolution and adoption in specific designs.
- **LO8**. Understand and select appropriately between various alternatives for data communications within a mechatronic system.

Critical Skills Development

- To plan workload and work to a deadline
- To identify and manage future risks
- To work effectively in a group
- To prototype hardware and software
- To have strong ability for analytical thinking and logical problem solving
- To debug effectively
- To use hand tools capably, including soldering
- To work and communicate at a standard approaching that of a professional engineer

Working in Groups

- Assignment 1 and 2 and Major Project require hardware
- The course has available hardware for 25 groups
- Please form groups of **min 4** and **max 5** students
- The total marks for group projects are **38%**

- At the end of the project, each group member will complete a peer and self evaluation form to assess the contributions of their teammates through SparkPlus.
- Peer and self evaluations will remain confidential and will be used as part of the individual contribution assessment.
- **SparkPlus completion is mandatory!!!**
- **Your final group mark will be scaled by the reported contributions.**
- **Students who do not complete SparkPlus risk 0 scaler!!!**



Working in Groups



- Establish communication channels (email, zoom, discord, slack)
- Projects will be challenging, divide the tasks properly, share responsibilities
- Plan your work on time
- Keep record of the progress and results
- Establish code style rules
- Backup your code!
- Bring the board to the lab always!
- **The teaching team will not intervene on how to collaborate!**
- You are entitled to leave the group if necessary. However, it is your responsibility to find a new team of up to five students and communicate the change to the teaching team.
- If you cannot find a team, we are unable to provide hardware.

Generative AI

- Using generative AI in mechatronic assignments can enhance both creativity and efficiency when done correctly.

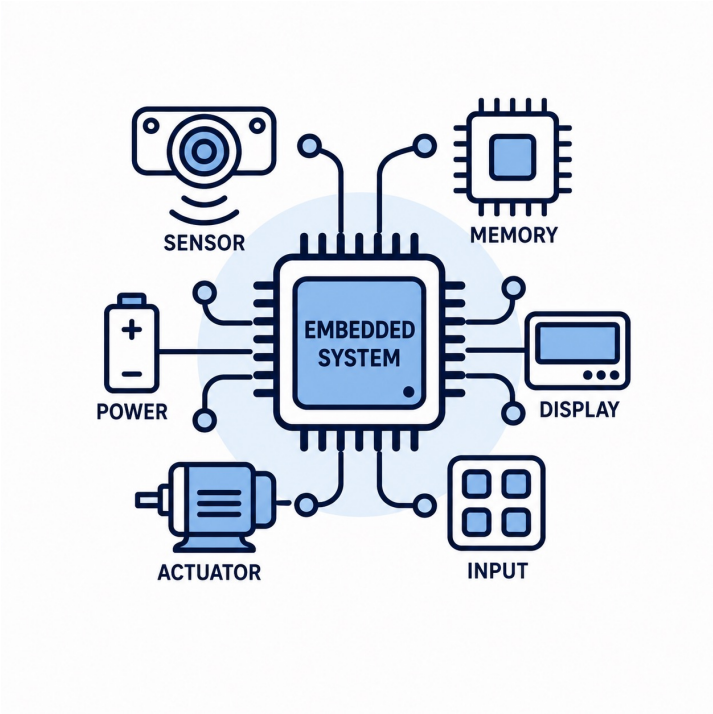
- Always ensure you thoroughly understand and verify the outputs to maintain academic integrity and accuracy.

- For coding tasks, AI can assist by providing code templates or debugging suggestions, but it's crucial to review and test the code to ensure it meets the specific requirements of your assignment.

- **In your report, you must always indicate when you use generative AI** by:
  a) attaching (in appendix) the AI generated solution and
  b) discussing the solution, your test and validation!

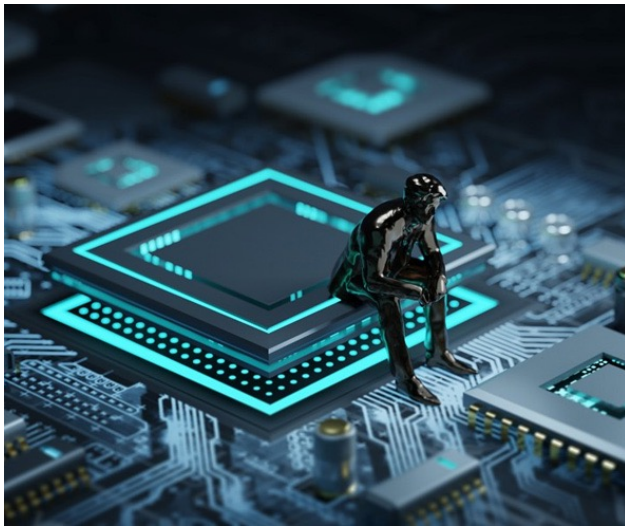
Learning Outcomes

- How to design and program **embedded systems** for mechatronics applications
- **Embedded systems:** Application-specific computer system which is built into a larger system or device
- Usually designed to execute a specific task
- Often runs dedicated software
- Often perform real time operations.



What are the Benefits of Embedded Systems?

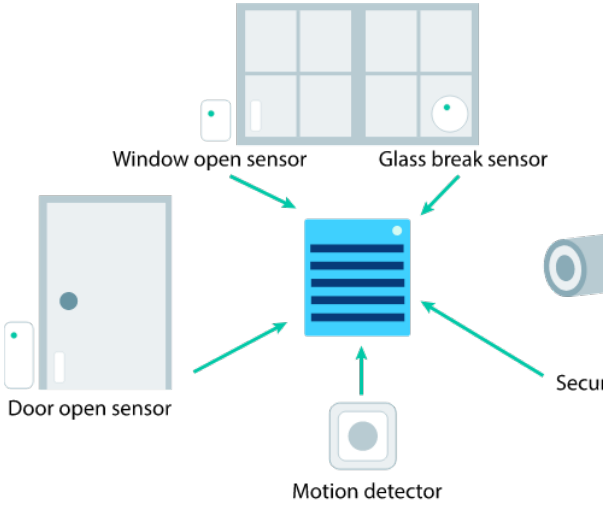
- Reduced cost: cheaper than analog solutions
- Increased functionality: allow constant monitoring of systems
- Improved performance: can operate independently for years
- Increased overall reliability: can provide immediate feedback



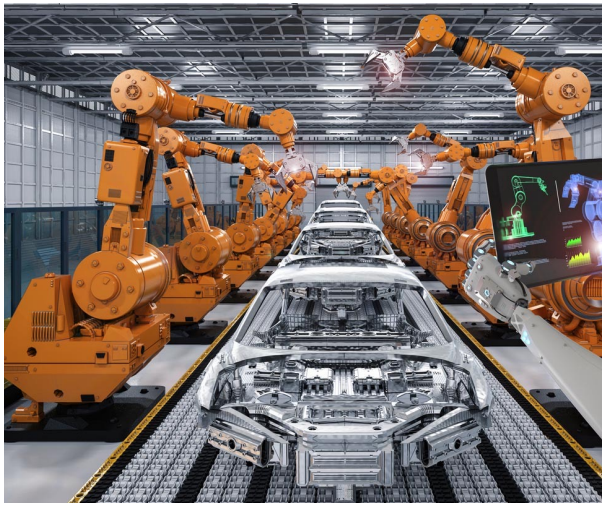
Examples of Mechatronic Embedded Systems

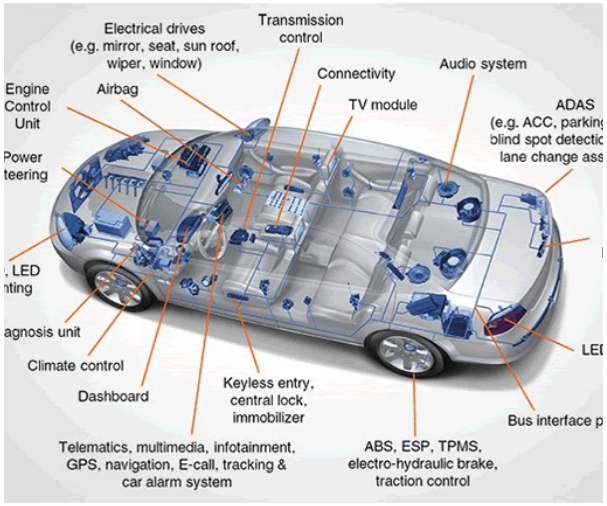
- Automotive Systems: specific modules inside the car: adaptive cruise control, smart airbags, engine checks, diagnostic info…
- Consumer electronics: mobile phones, videogame consoles, digital cameras, DVD players and GPS receivers
- Industry: control unit sending signals to correction elements such as pumps, valves, heaters or motors, etc. or sensors sending data back to control units
- Robotics: terrestrial, underwater, humanoids, space, advanced manufacturing

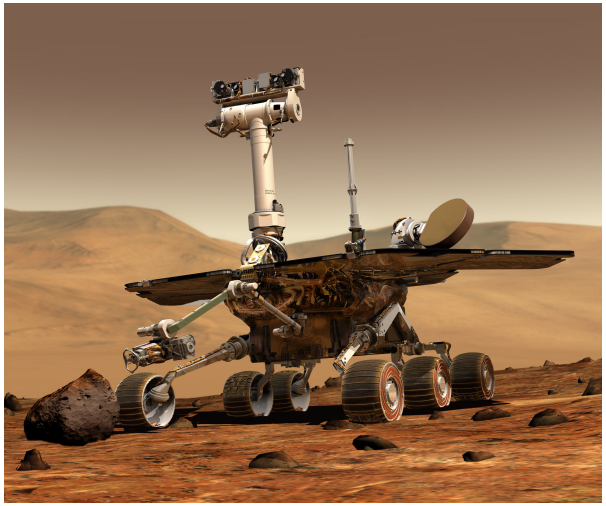



Window open sensor
Glass break sensor
Door open sensor
Motion detector






Electrical drives (e.g. mirror, seat, sun roof, wiper, window)
Transmission control
Connectivity
Audio system
Engine Control Unit
Airbag
TV module
ADAS
Power
LED
Climate control
Dashboard
Keyless entry, central lock, immobilizer
Telematics, multimedia, infotainment, GPS, navigation, E-call, tracking & car alarm system
ABS, ESP, TPMS, electro-hydraulic brake, traction control



Conclusions

- Learn how work and communicate at a standard approaching that of a **professional engineer**
- All assignments will be **group assignments**
- Ed Lessons are **individual assessments**
- **Embedded systems** are application-specific computers built into a larger system or device
- They improve the performance, functionality and features while lowering the cost and increasing the reliability of a system